Sai Vivek Peddi

Objective: Machine Learning Engineer

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WORK EXPERIENCE

JP Morgan Chase — Palo Alto, CA
Senior Machine Learning Engineer — Jan '23 - Ongoing

- Deployed **OpenHands** with **LiteLLM proxy**, enabling autonomous agents that can write code, run commands, and browse.
- Launched **AI Code Studio** with **EKS-hosted** sandboxes, **AWS Bedrock** endpoints, and **JPMC MCPs**, serving **1000+** devs.
- Built **MCP** servers with **SSO** auth and vector DBs for semantic search, adapting coding agents to JPMC specific toolchains.
- Built telemetry and observability infra using **Datadog** and **Splunk**, monitoring **1M+** daily requests with **99.9%** uptime.
- Implemented distributed Balance Loader (**4,000 TPS**) and orchestrated cross-region jobs via **Kafka** based on S3 events.
- Engineered **multi-threaded** S3 file processor with partitioned readers, reducing I/O operations by **5x** in production.

KFintech — Hyderabad, India
Solutions Architect & Software Engineer — Feb '19 - July '21

- Architected **Digix** a reporting and data visualization tool for **18** AMCs with **$137B** AUM and **100M** investor accounts.
- Led a team of **5** to write large scale data transformation Jobs in **Spark**. Reduced query times by **40** fold with new pipelines.

EDUCATION

University of California, Davis — Davis, CA
Masters degree in Computer Science (GPA: 4/4) — Sep '21 - Dec '22

- **Courses:** Adv Deep Learning, ML & Node Discovery, Modern Parallel Architecture, Distributed Database Systems, Quantum Simulations, Adv Visualization, Computer Architecture, Computer Networks, Operating Systems.
- **TA:** ML - *Fall 2021*, *Spring 2022* **,** Programming Languages - *Winter 2022* **,** Design & Analysis of Algorithms - *Summer 2022*

Birla Institute of Technology and Science (BITS), Pilani — Hyderabad, India
Bachelor of Engineering (Hons.) in Electronics & Communication Engineering (GPA: 7.89/10) — Aug '15 - Dec '18

IIIT Hyderabad — Hyderabad, India
Research Assistant, ML & Speech Processing Lab — Aug '18 - Jan '19

TECHNICAL SKILLS

- **Programming:** Python, Java, C, C++, JS, Cuda, SQL. • **ML:** PyTorch, tf, CuDNN, NCCL, ROCm, Spark, Hadoop, Hive, Kafka.
- **RAG:** LangChain, LangGraph • **Other:** AWS(EMR, Athena, RDS, S3, SageMaker), Terraform, pandas, Spring, git, docker, K8s.

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** : Credential, **Lang Chain Academy - Intro to Langgraph** : Credential

PROJECT HIGHLIGHTS

ML Pal - Multi-Agent RAG System Framework — June '24 - Ongoing

- Built deterministic multi-agent RAG systems using **LangGraph** and **LangChain**, with structured outputs and tool calls.
- Developed async persistence layers with **PSQL** for agent workflow state management, checkpoints, and long-term memory.
- Created enterprise-grade AI agents like **AWS Metrics Agent** with NL-to-SQL conversion and vector-based similarity search.

LLM Hardware Acceleration Suite — Aug'23 - Ongoing

- Leading a team of **15 CS grad** students in developing hardware accelerators for **LLMs**, in partnership with **AMD** researchers.
- Conducting research on **Heterogeneous Training and Inference**, focusing on cross-vendor GPU compatibility in **ML tasks**.
- Researching **DPU-based Smart NICs** for LLM task offloading, aimed at enhancing network and processor efficiency.

Multi-head attention with Sparse GPU kernels — Oct '22 - Dec'22

- Researched parallelism and sparsity in the **multi-head attention** module inside each encoder of a **Vision Transformer.**
- Developed the CUDA kernels for Self-attention using **cuSparse SPMM** and **cuBLAS SPMM** and experimented on **1660 Ti**.
- Able to achieve a **100x** speedup with **85%** sparsity and **97.5%** compute capacity of the GPU making it arithmetic bound.

Leveraging network delay variability to improve QoE of Latency Critical (LC) services — Jul '22 - Sep '22

- Engineered **K8s modules** with **Control Plane** for cloud scaling using Prometheus telemetry, achieving End-to-end SLOs.
- Able to guarantee a QoE target for more than **75%** utilization using EDF (**15%** higher than FCFS) for every pod in the cluster.

Application of Transformers in Audio Classification (Paper) (Code) — Mar '22 - Jun '22

- Built **self-supervised** models for Audio classification using transformers to tackle **inductive bias** from the existing CNNs.
- Experimented **ViT**, **MAE**, **DeIT** and **Swin Transformers** on **Mel-Spectograms** extracted from audio clips. Beat SOTA by 8%.

Privacy First Query Optimized Horizontal Partitioning using Machine Learning (Code) — Jan '22 - Mar '22

- Developed an ML model using **KModes** to build partitions automatically based on past run queries.
- Achieved up to **3x** speedup using **vertical partitioning** and up to **9x** speedup using **horizontal partitioning**.

Volumetric Isosurface Rendering with Deep Learning-based Super-Resolution (Video) — Sep '21 - Dec '21

- Developed an advanced visualization system to interact with **3D Volumetric Data** sets like CT Scans & Super Novas.